



Medicolegal implications of the Royal College of Radiologists standards for the interpretation and reporting of imaging investigations: a retrospective review of claims in abdominal radiology

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AIM: United Kingdom medicolegal claims are rising. Many radiological allegations cite failure to report incidental findings, but these are often inconspicuous and asymptomatic and therefore difficult to detect. Nevertheless, claimants often support their allegations by citing the Royal College of Radiologists standards for the interpretation and reporting of imaging investigations. We examined our medicolegal practice to determine the proportion where allegations cited incidental pathology.

METHODS: Five medicolegal expert witnesses, working in abdominopelvic radiology, reviewed their medicolegal practice and determined the proportion of recent cases in which allegations of substandard reporting cited failure to diagnose incidental pathology.

RESULTS: We identified 112 individual cases, comprising 176 imaging examinations; 99 (56 %) were computed tomography and 36 (20 %) magnetic resonance imaging. One hundred forty-four (82 %) were reported in the National Health Service, 14 (8 %) outsourced, and 18 (10 %) private practice. Medicolegal allegations cited failure to report incidental pathology in 41 (37 %) of the 112 patients. The commonest allegation in this group was failure to identify tumour (23 of 41 patients, 56 %). Interpretation was judged as substandard in 12 (29 %) of these patients compared with 23 (32 %) of the 71 patients in whom allegations did not cite incidental findings.

CONCLUSION: A substantial proportion of medicolegal allegations in abdominal radiology cite failure to report incidental pathology. College standards should recognise that unanticipated pathology is sometimes difficult to detect, especially without a clear steer from the referring clinician.

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Introduction

United Kingdom medicolegal claims continue to rise steeply. A 2021 analysis found that the total annual cost of National Health Service (NHS) litigation had reached £3.6 billion for 2018/2019.¹ Unfortunately, clinical radiology is well represented: of 10,900 NHS clinical negligence claims received for the year 2023/2024, 4.6 % were radiological, exactly the same total as for general medicine.² Because medical imaging is almost always archived, it is simple to establish whether an abnormality was present or not at the time of the original interpretation. Most radiological claims allege “missed” pathology.³ The Royal College of Radiologists “standards for interpretation and reporting of imaging investigations” (Second Edition, March 2018) state that, “images should also be specifically and methodically interrogated”, and that this may disclose, “abnormal findings both anticipated and unanticipated”.⁴ Unanticipated pathology is often described as an “incidentaloma”, defined by a former college president as, “any finding not directly related to the primary purpose for which the imaging examination was undertaken”.⁵ The authors all act as medicolegal expert witnesses, providing reports on liability for both claimant and defendant. We speculated that the proportion of cases based on failure to identify unanticipated incidentalomas was significant currently. Accordingly, we performed a retrospective review of our cases in order to determine the proportion where allegations cited failure to report unanticipated pathology.

Materials and methods

Ethical permission was not required because our study was a retrospective case series review. No patients were involved directly and there was neither intervention nor change in patient management. Information was anonymised before sharing so that no patient identifiable data passed among study authors.

We performed a retrospective review of our medicolegal practice. All authors searched their personal medicolegal records for cases where the primary radiological allegation centred on substandard reporting of abdomino-pelvic radiology (which is the primary subspeciality interest of all authors). We searched retrospectively from April 2024 until each individual author had accrued 30 consecutive cases or all available cases were exhausted, whichever occurred soonest. The limit of 30 cases was imposed, after discussion, to reduce individual burden since some experts had hundreds of individual cases. To avoid “double-counting”, any mutual cases (ie, where one expert was acting for the claimant and the other for the defendant, for

the same case) were allocated to the expert called on behalf of the claimant. Knowledge of mutual cases was identified via prior case conference between the experts (occurring as part of the claim) and not via data sharing.

We then recorded the following data: claimants’ age and sex at the time of the alleged radiological malpractice (at the time of index imaging since some claims involved multiple, sequential examinations); the imaging modality involved; the clinical indication for imaging stated on the imaging request; the specific allegation(s) made by the claimant; whether imaging was acquired in the NHS, outsourced by the NHS, or private sector; whether the expert was instructed by the defendant or claimant. The expert then noted whether the primary allegation involved an “unanticipated” finding or not and whether he considered the allegation of substandard reporting could be upheld. We defined an “unanticipated” finding as, “any finding not directly related to the primary purpose for which the imaging examination was undertaken”.⁵ So, for example, alleged failure to report recurrent tumour in a patient imaged for colorectal cancer surveillance would not be considered incidental, whereas alleged failure to report a renal tumour in the same patient would. Similarly, if the patient had a history of colorectal cancer but this was not stated in the clinical details provided to the radiologist and the patient was being imaged for an alternative indication, then failure to report colorectal recurrence would count as an “incidental” case. Where multiple imaging episodes and/or modalities were performed in an individual patient, we included all instances where the expert was required to respond to a claimant’s allegation of substandard reporting. We excluded any case where there was no allegation of substandard reporting (eg, expert instructed to determine tumour stage, extent of injury etc.).

The first author assembled the data and categorised the clinical indication for each individual imaging investigation into broad groups, identified by thematic analysis.

The first author also categorised the primary medicolegal allegation for each case into 11 individual categories, again identified by thematic analysis. Data were analysed using simple frequencies and proportions. Ninety-five percent confidence intervals around percentages were calculated using the binomial exact method (STATA 18.0, StataCorp).

Results

Five experts contributed the data. Their medicolegal experience (years) was 3, 20, 10, 13, and 24 at the time of data collection (April 2024), respectively. They provided 10, 12, 30, 30, and 30 individual cases, respectively, totalling

112 patient cases overall (median age: 53 years, range: 14–81 years; 37 male, 75 female). The claimant to defendant split was 92 (82 %) to 20 (18 %).

The experts contributed data on 12, 17, 41, 47, and 59 individual imaging examinations respectively, totalling 176 examinations, acquired 2001 to 2023 inclusive. These comprised 99 (56 %) computed tomography (CT; including CT colonography), 36 (20 %) magnetic resonance imaging (MRI; including small bowel and magnetic resonance cholangiopancreatography (MRCP)), 16 ultrasound (9 %), 13 (7 %) plain film, and 12 (7 %) fluoroscopic examinations. The individual number of examinations requiring a medico-legal opinion per case ranged from 1 to 12, median 1. Of the 176 imaging investigations, 144 (82 %) were reported in the NHS, 14 (8 %) were outsourced by the NHS, and 18 (10 %, from 9 patients) were reported in private practice. Broad categories describing the clinical indication for each individual imaging investigation were as follows: the commonest was “abdominal symptoms” (99 examinations, 56 %), followed by “postoperative complication” (21 examinations, 12 %), “tumour recurrence/surveillance” (18 examinations, 10 %), “? Sepsis” (17 examinations, 10 %), and “weight loss” (5 examinations, 3 %), followed by various miscellaneous indications.

Broad categories describing the primary medicolegal allegation for each case were as follows: the commonest was “primary tumour unreported” (n=40 cases, 36 %), followed by “perforation unreported” (n=12, 11 %), and “obstruction unreported” (n=9, 8 %; Table 1. In 41 (37 %, 95%CI: 28 %–46 %) of the 112 patients, the experts judged the pathology relating to the allegation as incidental to the clinical indication for imaging (Table 1). Of these 41 patients (comprising 50 individual examinations), the original interpretation was judged to be substandard in 12 (29 %, 95%CI: 16 %–46 %) patients (12 examinations). In the remaining 71 patients (63 %, 95%CI 54 %–72 %), the experts

judged the pathology relating to the allegation as not incidental. Of these 71 patients (126 examinations), the original interpretation was judged to be substandard in 23 (32 %, 95%CI: 22 %–45 %) patients (37 examinations). There was no significant difference between the proportion of allegations judged substandard when patients with incidental and non-incidental findings were compared (29 % vs 32 %, chi-squared: 0.12, $P=0.73$).

Overall details of the 29 patients in whom interpretation of an incidental finding was judged reasonable (ie, not substandard) are given in Table 2. Examples of interpretations of incidental pathology judged reasonable included failure to report a small pancreatic tumour in a patient undergoing dedicated small bowel MRI to search for Crohn’s disease; failure to report a small renal tumour in a patient undergoing lumbar MRI to investigate back pain; and several examples of failure to diagnose small colonic tumours (including flat polyps) in patients undergoing standard abdominopelvic CT, ie, where there had been no bowel preparation or gas insufflation.

Discussion

Radiological “incidentalomas” may be clinically significant (eg, lung tumour) or insignificant (lung granuloma) and judged easily perceptible (large tumour) or not (small tumour), and have provoked medicolegal action for decades.⁶ Of course, error is inevitable for all practicing radiologists,^{7,8} so “missing” pathology does not automatically imply substandard (ie, negligent) practice. In the UK, the “Bolam test” is used to judge whether medical practice (which, for clinical radiology, usually means interpretation) is “reasonable” or “substandard”. Bolam considers whether a hypothetical body of responsible professionals (ie, “competent” and acting with “ordinary skill”) would act similarly.⁹ This means that if the radiologist’s report was in accordance with the likely interpretation obtained from a body of radiologists, then it will not be found negligent. “In accordance” usually means majority opinion, although this may not apply in some areas where medical opinion diverges substantially. Expert medicolegal radiologists must therefore imagine themselves to be a properly trained “general” consultant radiologist faced with the clinical scenario at the time of the original interpretation and then come to a decision as to whether the issued report was substandard or not.

Hulson analysed 791 claims settled by the NHS Litigation Authority between 1995 and 2014, where the primary allegation focused on radiology.¹⁰ Almost one-third of claims related to missed or delayed tumour diagnosis but data were not granular enough to determine whether these were incidental to the primary clinical question posed to the radiologist. We found that failure to report primary or recurrent tumour was the primary allegation in nearly half of our patients, with the tumour judged an incidental finding in more than half of these. Overall, our review found that nearly 40 % of allegations cited pathology that

Table 1

Primary medicolegal allegation in 112 patients, split by whether the allegation was judged incidental to the clinical indication for imaging (n=41) or not (n=71).

Medicolegal allegation	Allegation not an incidental finding (n=71)	Allegation an incidental finding (n=41)
Primary tumour unreported	17	23
Perforation unreported	9	3
Obstruction unreported	8	1
Recurrent tumour unreported	7	4
Sepsis unreported	7	0
Bariatric complication unreported	6	1
Cancer mis-staged	3	0
Thrombus unreported	3	2
Bowel ischaemia unreported	2	0
Stone unreported	1	2
Miscellaneous	8	5

Table 2
Details of 29 patients in whom interpretation of an incidental finding was judged reasonable.

Case	Source	Imaging modality	Clinical circumstances	Allegation
1	NHS	CT	Renal cancer surveillance	Colon cancer unreported (unprepared study)
2	Outsourced	MRI small bowel & CT colonography	? Crohn's? Colonic malignancy.	Pancreatic cancer unreported
3	NHS	CT	Anaemia	Gastric varices unreported
4	NHS	CT	Abdominal pain? Cause	Volvulus unreported (not obstructed at the time of scanning)
5	NHS	CT	Pancreatic cancer suspected.	Hepatic tumour unreported
6	NHS	CT	Left iliac fossa pain? Diverticulitis	Right-sided colon cancer unreported (unprepared study)
7	NHS	Small bowel MRI & CT	Assess Crohn's disease activity	Small bowel tumour unreported
8	NHS	CT	? Postoperative collection.	SMA thrombus unreported
9	NHS	CT	Prostate cancer surveillance	Ureteric tumour unreported
10	NHS	MRI	? Anal fistula.	Rectal cancer unreported
11	NHS	CT, MRI, fluoroscopy	Anastomotic leak from bowel?	Hydronephrosis unreported
12	NHS	Abdominal film	? Bowel obstruction	Pelvic mass unreported
13	NHS	CT and US	Renal stone assessment	Colorectal cancer recurrence unreported
14	Outsourced	US	? Gallstones	Liver metastases unreported
15	NHS	CT	Abdominal pain ? Cause	Flat colonic polyp unreported (unprepared study)
16	NHS	CT	? Lymphoma	Deep vein thrombosis unreported
17	NHS	CT	? Postoperative collection	Peritoneal tumour recurrence unreported
18	NHS	CT	Pain following pancreatectomy	Flat colonic polyp unreported (unprepared study)
19	NHS	CT	? Appendicitis	Coeliac axis stenosis unreported
20	NHS	MRI	Back pain	Renal cancer unreported (lumbar spine MRI)
21	NHS	CT colonography	Constipation	Pancreatic cancer unreported (non-contrast scan)
22	NHS	MRI	Constipation	Suprapubic catheter in caecum unreported (evacuating proctogram)
23	NHS	CXR	Abdominal pain? Cause	Diaphragmatic hernia unreported
24	Outsourced	CT	Anaemia	Gastric cancer unreported
25	NHS	MRCP	Abdominal pain? Cause	Displaced Filshie clips unreported
26	NHS	CT	Colonic cancer surveillance	Terminal ileal Crohn's disease unreported
27	NHS	CT	Change in bowel habit	Synchronous colonic cancer unreported (unprepared study).
28	NHS	CT and fluoroscopy	? Postoperative collection/leak	Roux-en-O misconfiguration unreported
29	NHS	CT and US	? Gallstones	Synchronous gallbladder cancer unreported

CT, computed tomography; MRI, magnetic resonance imaging; NHS, National Health Service; SMA, Superior mesenteric artery; MRCP, Magnetic resonance cholangiopancreatography; US, Ultrasound; CXR, Chest x-ray.

was considered incidental to the primary clinical question being asked of the radiologist.

While an expert witness will consider some incidentalomas “unmissable” by any competent radiologist when applying the Bolam test, others will be judged missable by the, “majority of a body of consultant radiologists acting reasonably” (to use legal terminology), unless the radiologist is given a specific steer (and perhaps not even then). Perhaps unsurprisingly, two meta-analyses have concluded that diagnostic accuracy increases when radiologists are provided with relevant clinical details.^{11,12} For example, we found several examples where the allegation cited failure to report tumours that were incidental to the clinical question being asked, often imaged using an inappropriate test (notably failure to report small colonic tumours from conventional abdominopelvic CT), and frequently where a prior history of malignancy was not supplied in the clinical details.

The 2018 Royal College standards make no distinction between an “unanticipated” finding sufficiently apparent that no reasonable radiologist should fail to report it, and an unanticipated finding that is so subtle it requires a dedicated and concerted search to unearth (such as that precipitated by a steer from the requesting clinician). It could be argued, therefore, that these college standards oblige radiologists (and reporting radiographers, sonographers, etc.) to detect and report all visible pathology, irrespective of the clinical details provided, and irrespective of conspicuity and clinical significance. This stance is one that all authors have encountered in their personal medicolegal practice, ie, the college guidelines are used by the claimant to support allegations of substandard practice, even when pathology is both incidental and inconspicuous.

A requirement to identify any pathology incidental to the clinical question and/or symptoms could be interpreted as screening. We note that the UK National Screening

Committee spends considerable resource deciding who should be screened, what for, when, and with what test. While organised screening programmes seek to diagnose specific pathology in well-characterised groups of attendees, modern medical imaging generates a plethora of data that encourages surreptitious screening for unanticipated, wide-ranging, asymptomatic pathology. This appears to burden radiologists specifically; for example, breast surgeons are not generally expected to detect incidental colonic cancers.

We believe it is unreasonable and unrealistic to expect radiologists to unfailingly identify incidental, subtle pathologies when the clinical details direct attention elsewhere; we discovered that allegations related to incidental pathology in 37 % of our cases. We are certainly not arguing for lowered standards that excuse inadequate concentration: radiologists, quite rightly, should be expected to diagnose relevant pathology, both anticipated and unanticipated. However, the issue is at which point does unanticipated pathology become imperceptible without a specific steer? This is especially the case in a healthcare system under intolerable pressure and where radiologists are increasingly presented with less-specific referrals. Clear differential diagnosis is frequently replaced by symptomatology alone; “abdominal pain ? Cause”. It is the authors’ experience that a large proportion of referrals now originate from non-medically qualified personnel, untrained in differential diagnosis.

The issue is exacerbated by the move in some legal circles to displace the Bolam test in favour of “pure diagnosis”, which conjectures that if pathology is at all visible, then failure to report it automatically represents substandard practice.⁹ We believe this reflects a lack of understanding on the part of some legal professionals that radiological diagnosis is subjective and therefore subject to individual variation when the clinical importance of a potential finding is considered. All authors of this article have experienced “fishing trip” instructions from referring solicitors, where they have been asked to interpret all available imaging, with the hope of identifying any substandard practice in the absence of a prior, firm allegation. All authors discourage this stance and take the position that experts require specific direction because, as we have outlined, clinical context and interpretation are inseparable.

Ultimately, we are not arguing for lower standards but believe that the spirit of the 2018 standards as written is being misinterpreted by some medicolegal experts (and/or solicitors/barristers/claimants). Indeed, all authors wish to emphasise the need for college standards and agree fully that it is right and proper that radiologists are regulated and held accountable. The general public also expect medical professions to be regulated. While the college may wish to avoid an explicit medicolegal stance, it is undeniable that the standards will be cited in court and, presently, they appear insufficiently granular regarding detection of anticipated vs unanticipated pathology. Without acknowledgement that incidentalomas pose specific

issues, some radiologists may adopt overly defensive reporting and/or devote undue amounts of time to interpretation where the clinical question is clear and precise. Just as the Bolam test stipulates “reasonable” practice, standards should state what constitutes reasonable interpretation in the light of the information and intent provided by referring clinicians.

Our review does have limitations. As a group of abdominal radiologists, our data are restricted to cases in this field. Accordingly, we cannot say if our findings generalise to other areas of imaging, although we would expect them to. The decision as to whether an allegation related to an incidental finding or not was made subjectively, by the expert instructed originally to handle the case; we wished to avoid transfer of patient identifiable data, so authors only searched their own cases. Because there is often a legal instruction to destroy case details on settlement, we did not have data for some cases at the time of our audit. When preparing this article, we were aware that the 2018 college standards were being revised and these were published¹³ subsequent to our article submission; one of our authors was a member of the guideline development group. The October 2025 standards state that the radiological report should, “allow incidental findings to be mentioned”, going on to state that, “there should be provision to ensure competent and holistic review of all relevant findings”, noting that, “responsibility for reviewing nontarget areas lies with both the reporter and the employer”.¹³ The new guidelines also state that, “interpretation may be incorrect if referral details are deficient, absent, or misleading”,¹³ which appears to potentially protect radiologists who miss subtle, incidental pathology without a specific steer. However, from a legal perspective, the standards to which radiologists are held are those in place at the time of the alleged substandard practice and not those most recently published.

In summary, a large proportion of medicolegal allegations in abdominal radiology cite failure to report pathology that is both inconspicuous and incidental to the clinical request, and many support their allegations by citing the 2018 Royal College of Radiologists Standards for the interpretation and reporting of imaging investigations. College standards should reflect the fact that unanticipated pathology is sometimes difficult to detect, especially when the clinical details provided direct radiological attention elsewhere.

Author contributions

SH contributed as guarantor of integrity of the entire study.

SH contributed to study concepts and design.

SH contributed to literature research.

All authors contributed to clinical studies.

SH contributed to experimental studies/data analysis.

SH contributed to statistical analysis.

SH contributed to manuscript preparation.

All authors contributed to manuscript editing.

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Conflict of interest

The authors declare the following financial interests/ personal relationships which may be considered as potential competing interests: Professor Steve Halligan reports financial support was provided by National Institute for Health and Care Research University College Hospitals Biomedical Research Centre. Dominic Blunt reports a relationship with Royal College of Radiologists that includes: board membership. All authors act as expert witnesses for both claimants and defendants. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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